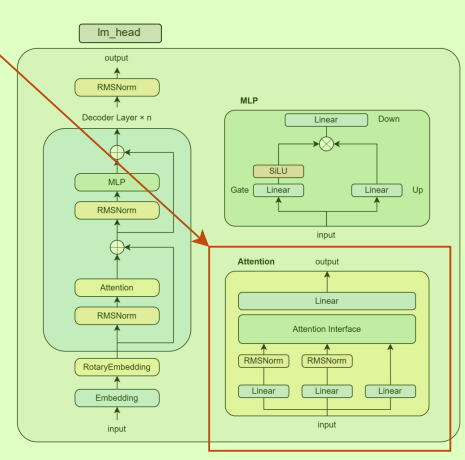


1.5 attention.py ²

```

Qwen3-72B结构
QwenForCausalLM
  (model): Qwen2LMHeadModel
    (embed_tokens): Embedding(51206, 1024)
    (layers): ModuleList(
      (0): nn.Sequential(
        (self_attn): Qwen2Attention
          (proj_in): Linear(in_features=1024, out_features=2048, bias=False)
          (proj_out): Linear(in_features=2048, out_features=1024, bias=False)
          (k_proj): Linear(in_features=1024, out_features=1024, bias=False)
          (v_proj): Linear(in_features=1024, out_features=1024, bias=False)
          (q_proj): Linear(in_features=1024, out_features=1024, bias=False)
          (k_norm): Qwen2RMSNorm(1024, eps=1e-06)
          (v_norm): Qwen2RMSNorm(1024, eps=1e-06)
          (gate_fn): GELU
        (ff): nn.Sequential(
          (proj_in): Linear(in_features=2048, out_features=3072, bias=False)
          (proj_out): Linear(in_features=3072, out_features=2048, bias=False)
          (act_fn): Linear(in_features=2048, out_features=2048, bias=False)
          (act_fn2): SiLUActivation
        )
      )
    (input_layernorm): Qwen2RMSNorm(1024, eps=1e-06)
    (post_attention_layernorm): Qwen2RMSNorm(1024, eps=1e-06)
    (norm): Qwen2RMSNorm(1024, eps=1e-06)
    (rotary_emb): Qwen2RotaryEmbedding
    (lm_head): Linear(in_features=1024, out_features=51206, bias=False)

```



Qwen3-0.6B
dtype = float32
4b

$$\frac{0.6 \times 1000000000 \times 4}{1024 \times 1024 \times 1024} = 2.23GB$$

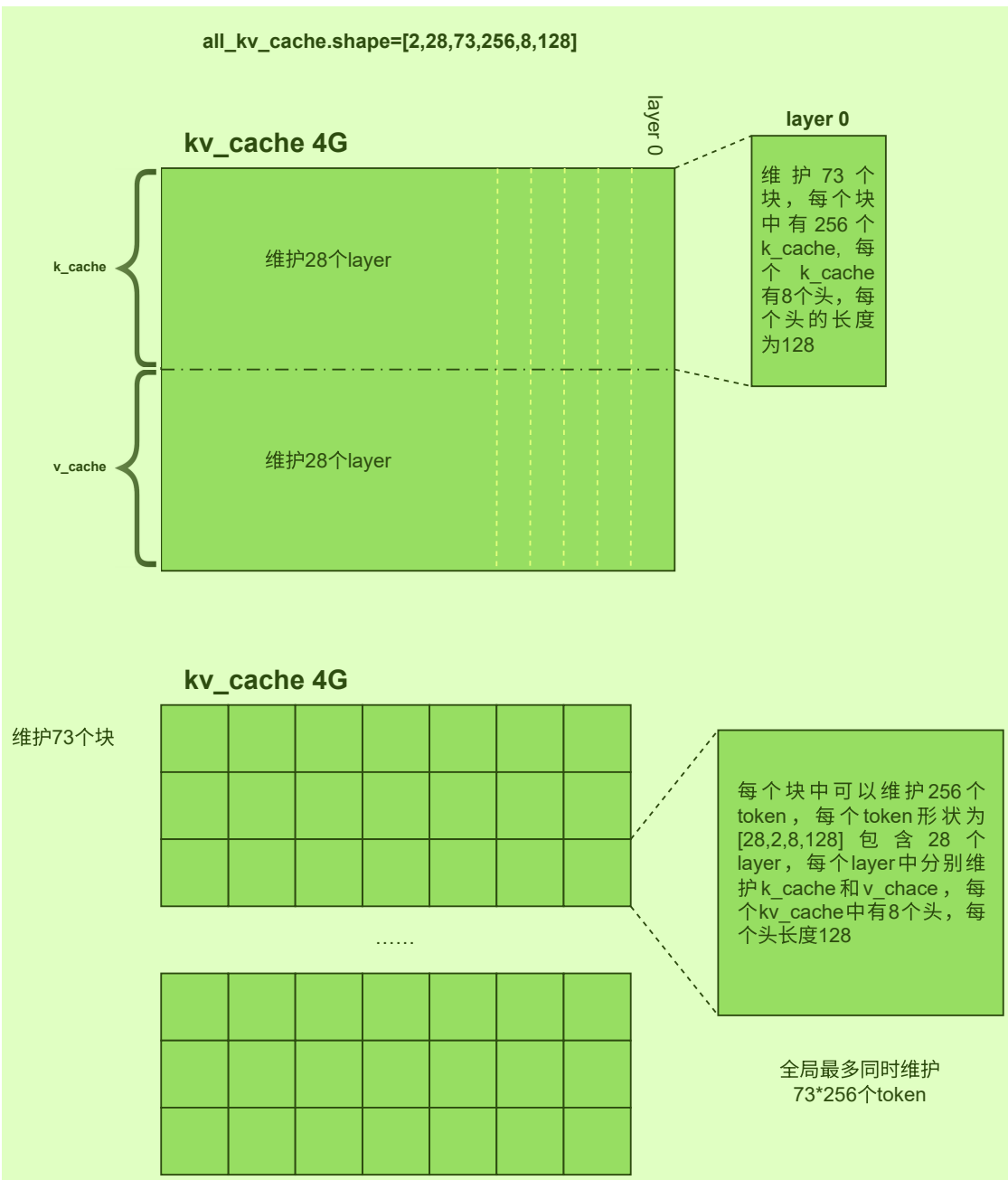
GPU HBM Memory 8G

Qwen3-0.6B Model 2.23 GB	可用于分配 kv_cache的空间 4 GB	预留空间
--------------------------------	------------------------------	------

$$\frac{4 \times 1024 \times 1024 \times 1024}{2 \times 28 \times 256 \times 8 \times 128 \times 4} = 73\text{块}$$

GPU HBM Memory 8G
全局可以维护73个kv_cache块

Qwen3-0.6B Model 2.23 GB	kv_cache 4 GB	预留空间
--------------------------------	------------------	------

[illegible]

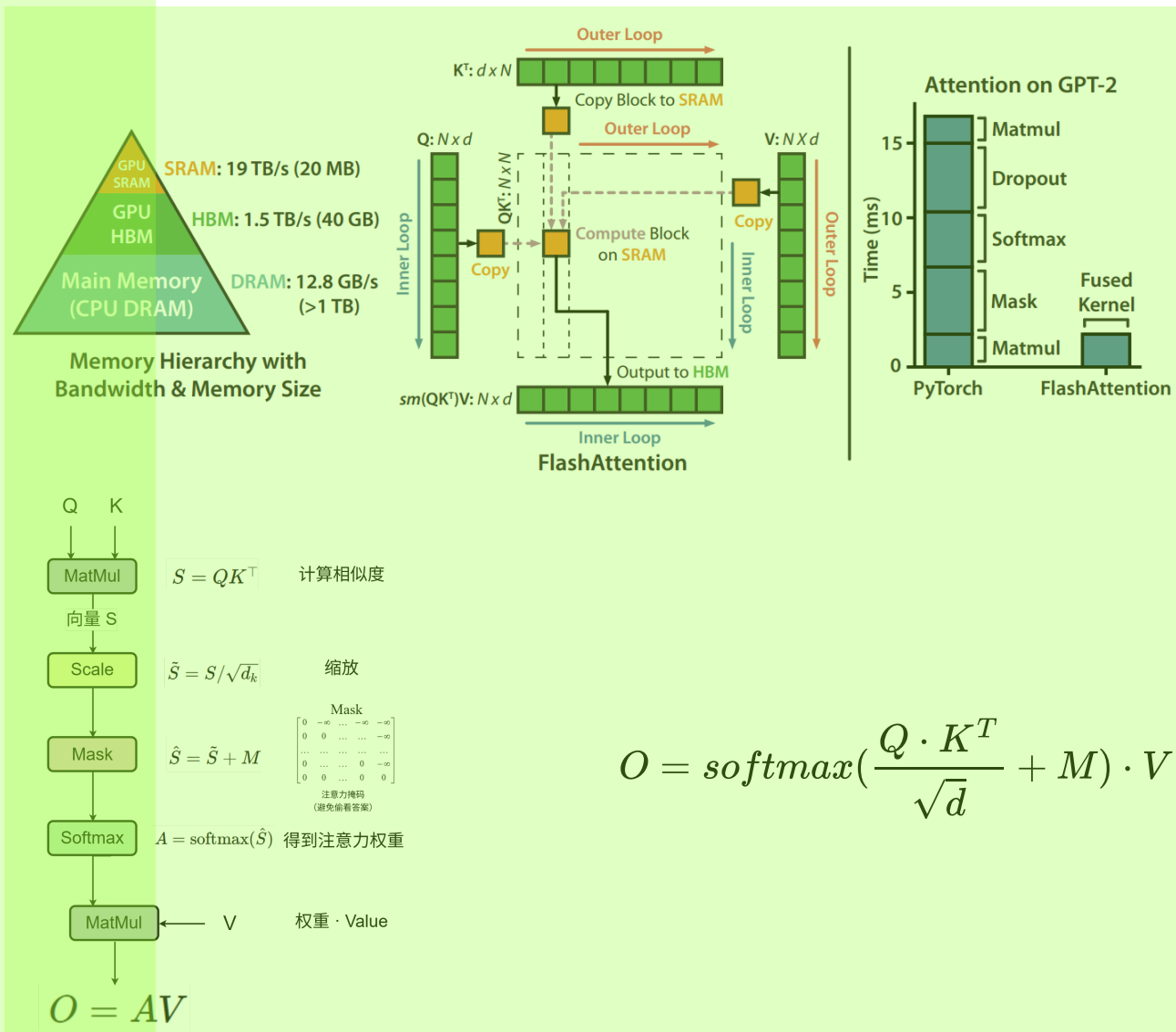
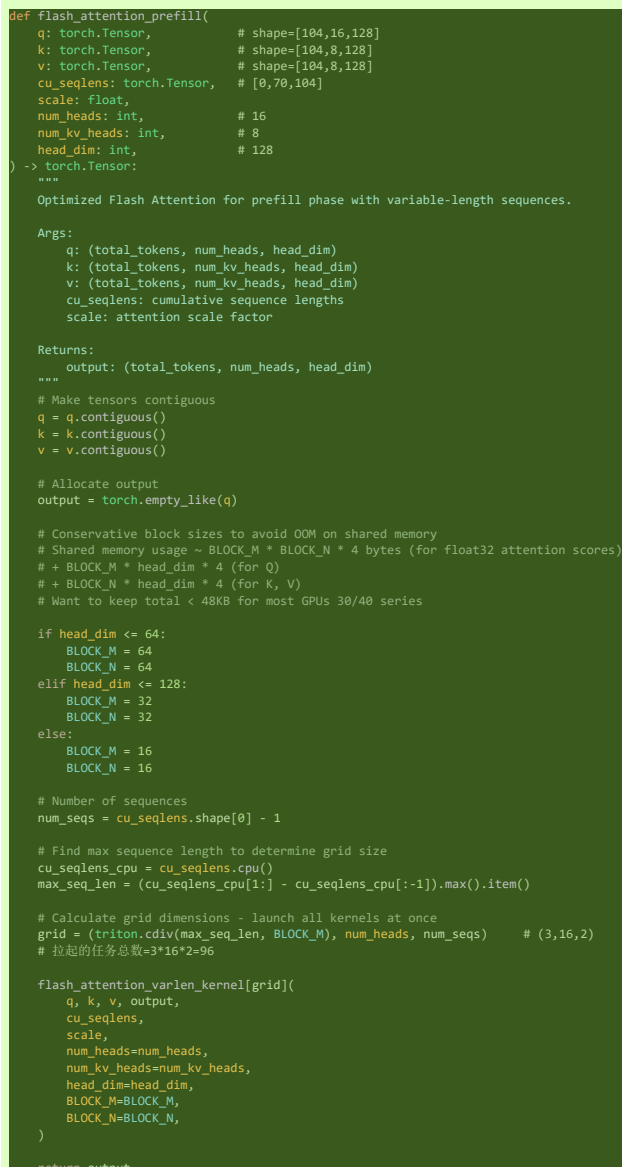
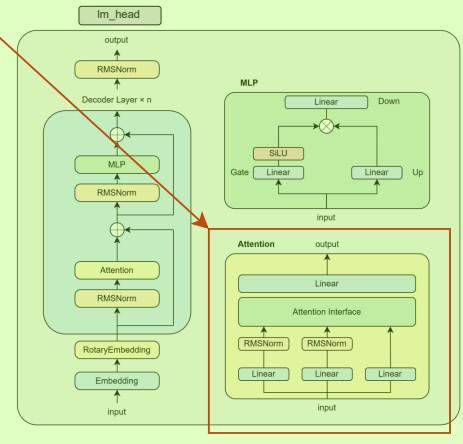
```

Qwen2-0.5B结构
(model) = Qwen2ForCausalLM(
  (embed_tokens): Embedding(51200, 1024)
  (layers): ModuleList(
    (0-27): 28 x Qwen2DecoderLayer(
      (self_attn): Qwen2Attention(
        (q_proj): Linear(in_features=1024, out_features=2048, bias=False)
        (k_proj): Linear(in_features=1024, out_features=2048, bias=False)
        (v_proj): Linear(in_features=1024, out_features=2048, bias=False)
        (o_proj): Linear(in_features=2048, out_features=1024, bias=False)
        (rotary_emb): Qwen2RotaryEmbedder(128, eps=1e-06)
      )
      (mlp): Qwen2MLP(
        (gate_proj): Linear(in_features=1024, out_features=3072, bias=False)
        (up_proj): Linear(in_features=1024, out_features=3072, bias=False)
        (down_proj): Linear(in_features=3072, out_features=1024, bias=False)
        (act_fn): SiluFunction()
      )
    )
  (input_layernorm): Qwen2RMSNorm(1024, eps=1e-06)
  (post_attention_layernorm): Qwen2RMSNorm(1024, eps=1e-06)
)

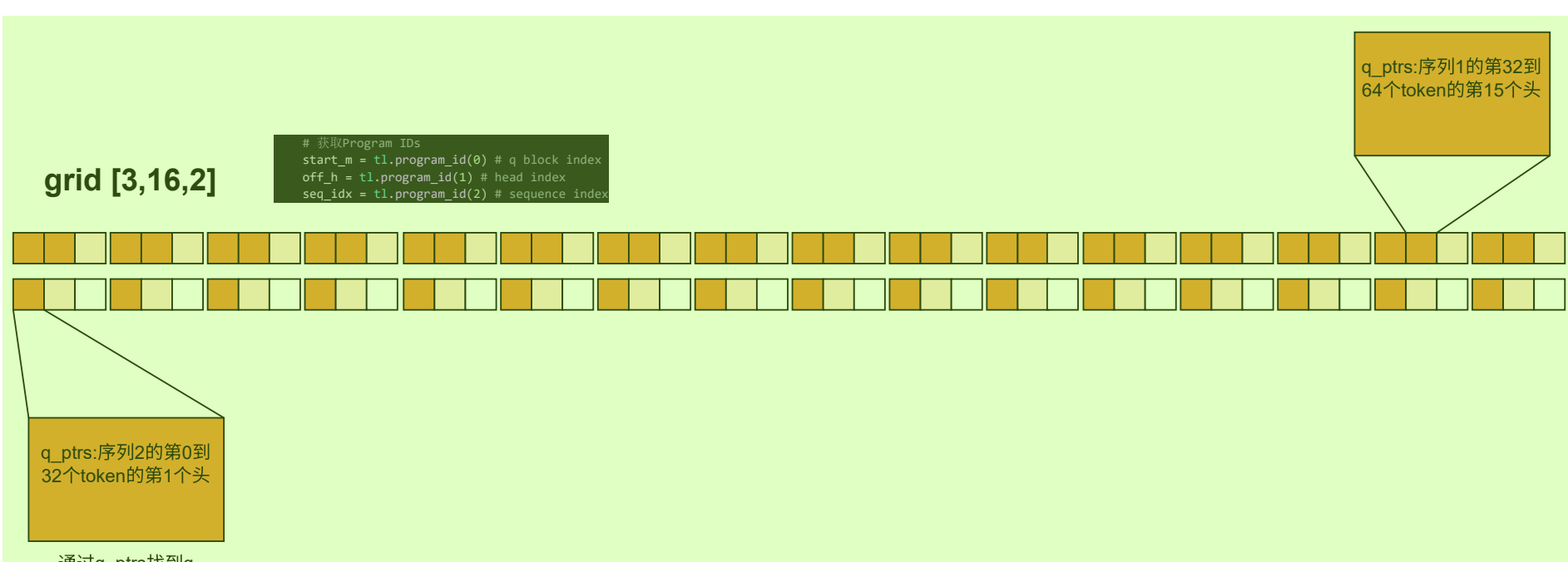
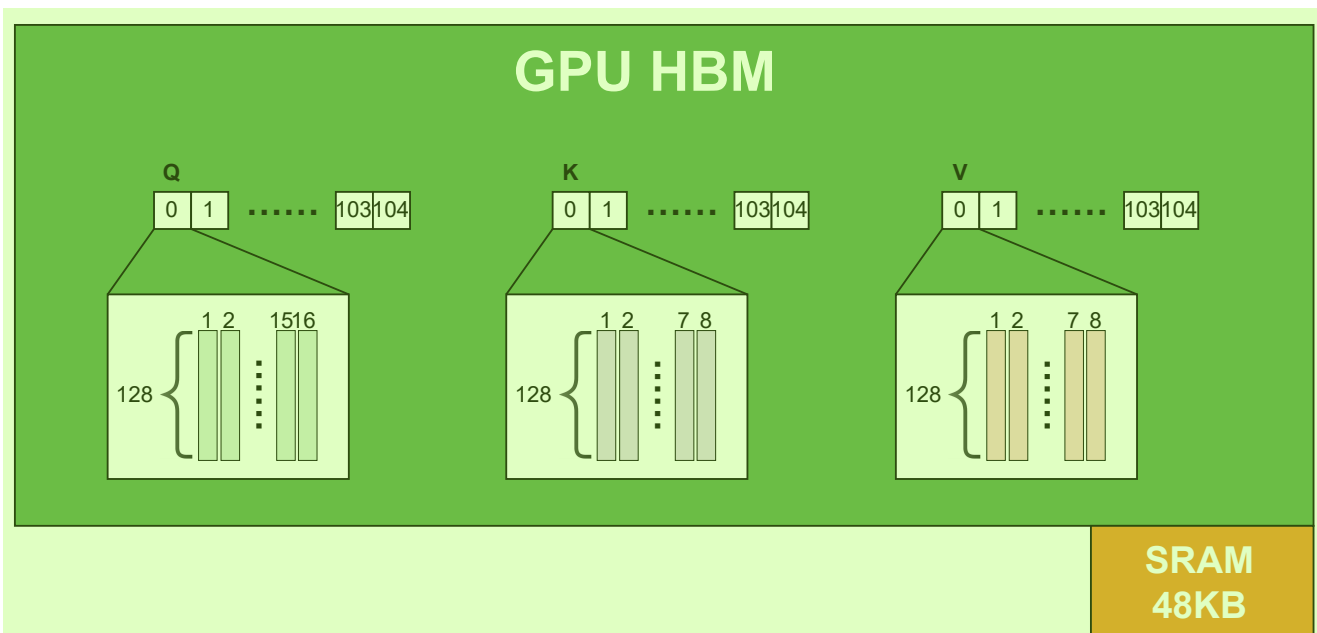
(norm): Qwen2RMSNorm(1024, eps=1e-06)
(rotary_emb): Qwen2RotaryEmbedder(128, eps=1e-06)

(ln_head): Linear(in_features=1024, out_features=51200, bias=False)

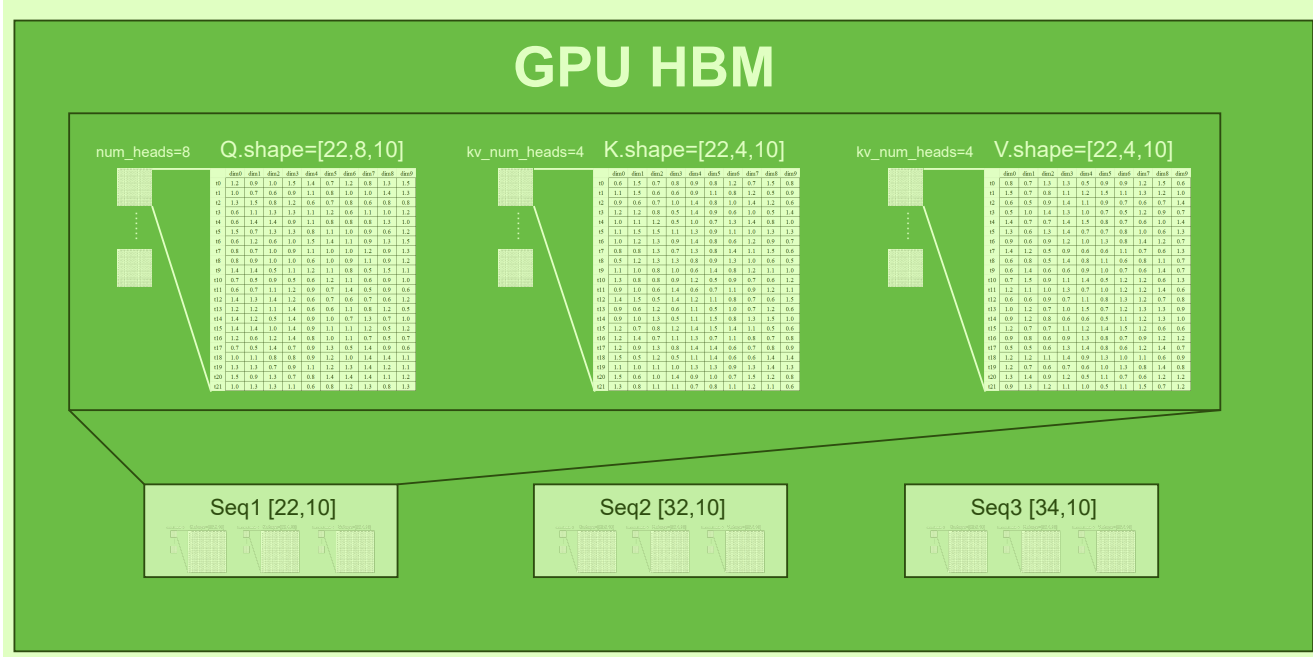
```



使用triton实现的FlashAttention¹²

[illegible]

计算流程（案例）¹



grid=[3,8,3]

